`oscfit`: An Automated Bayesian Framework and Seismological Inversion Pipeline for Decayless Kink Oscillations in Solar Coronal Loops

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Abstract

Context. The heating of the solar corona to mega-Kelvin temperatures remains a pivotal open question in solar-terrestrial physics. Decayless transverse kink oscillations in coronal magnetic loops represent a ubiquitous, continuous source of magnetohydrodynamic (MHD) wave energy capable of driving non-thermal resonant dissipation. However, systematic observational studies have been severely restricted by the obsolescence of legacy Interactive Data Language (IDL) tools (e.g., `Auto-NUWT`, `SoBAT`) and a lack of standardized, automated Python pipelines.

Aims. We present `oscfit`, an open-source, modular Python package designed for the end-to-end extraction, automated sub-pixel tracking, Bayesian model selection, and MHD seismological inversion of standing kink oscillations in solar extreme-ultraviolet (EUV) imaging data.

Methods. `oscfit` automates perpendicular slit extraction from 3D FITS data cubes, sub-pixel Gaussian spatial tracking, Lomb-Scargle and Morlet wavelet spectral significance estimation against red-noise baselines, affine-invariant Markov Chain Monte Carlo (MCMC) parameter estimation across six candidate physical models, and Bayesian evidence calculation via dynamic nested sampling. It directly derives the kink speed (C_k), internal phase speed (V_A), magnetic field strength (B), and wave energy flux (F) with rigorous analytical and Monte Carlo error propagation.

Results. We validate `oscfit` against three independent published catalogs: (i) the Solar Orbiter/EUI catalog of 105 active-region and 42 quiet-Sun loops (Shrivastav et al. 2024), where `oscfit` reproduces published kink speeds to 0.1% and magnetic field values to 0.5%, while uncovering an unstated internal density assumption in the published literature; (ii) the 221-event SDO/AIA decaying oscillation catalog (Nechaeva et al. 2019), achieving full distribution consistency with the published median (1407 km s^{-1} calculated vs. 1330 km s^{-1} published); and (iii) the SDO/AIA decayless baseline catalog (Anfinogentov et al. 2015).

Conclusions. `oscfit` establishes an open, reproducible, and standardized computational foundation for coronal seismology in the era of high-resolution space missions including Solar Orbiter/EUI, SDO/AIA, and Aditya-L1/VELC.

Keywords: Sun: corona — Sun: magnetic fields — Magnetohydrodynamics (MHD) — Waves — Methods: data analysis — Methods: statistical

1 Introduction

The heating of the solar corona to temperatures exceeding 1–3 MK and the continuous acceleration of the solar wind represent longstanding challenges in contemporary astrophysics

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[Aschwanden et al., 2002, Van Doorsselaere et al., 2008]. Among the prominent proposed heating mechanisms, magnetohydrodynamic (MHD) wave dissipation occupies a central position. In particular, transverse kink oscillations of coronal flux tubes (with azimuthal wavenumber $m = 1$) provide both a direct probe of local plasma and magnetic properties via coronal seismology and an active conduit for non-thermal energy transport [Nakariakov and Ofman, 2001].

Historically, observations of coronal loop kink oscillations were primarily confined to large-amplitude, flare-induced events exhibiting rapid, exponential damping [Aschwanden et al., 2002, Nechaeva et al., 2019]. The damping in these energetic events is commonly attributed to resonant absorption, where collective kink wave energy is efficiently transferred to torsional Alfvén waves within a thin, inhomogeneous boundary layer [Van Doorsselaere et al., 2008].

However, following the launch of the Solar Dynamics Observatory (SDO) and the High Resolution Imagers on board Solar Orbiter (EUI/HRI), high-cadence extreme ultraviolet (EUV) observations revealed a qualitatively distinct regime: *decayless* transverse oscillations [Wang et al., 2012, Anfinogentov et al., 2013, 2015, Shrivastav et al., 2024]. Unlike their flare-induced decaying counterparts, decayless kink oscillations:

1. Persist indefinitely throughout quiescent intervals without apparent decay;
2. Possess small transverse displacement amplitudes ($A \sim 0.05$–0.5 Mm);
3. Exhibit velocity amplitudes ($V \sim 1$–20 km s^{-1}) that are sub-resolution for standard imaging spectrometers yet resolved by ultra-fine EUV imagers;
4. Are ubiquitously present across both active region (AR) loops and quiet Sun (QS) multi-thermal loop complexes.

Despite their profound physical implications, extracting and analyzing coronal loop oscillations has historically been fraught with methodological hurdles. In over a decade of SDO operations, only approximately 223 decaying oscillations and fewer than 150 decayless loops were catalogued in the literature [Nechaeva et al., 2019, Shrivastav et al., 2024]. The primary technological bottleneck has been the reliance on proprietary, unmaintained software. For over a decade, loop coordinate tracking relied on the IDL package `Auto-NUWT` (unmaintained since 2019), while Bayesian fitting relied on `SoBAT` (inactive since 2020). Consequently, research groups have routinely developed bespoke, unvetted scripts with divergent detrending methods, undocumented noise treatments, and non-standardized error propagation, introducing an acute reproducibility crisis.

To overcome these barriers, we introduce `oscfit`, an open-source, fully automated, and peer-reviewed Python pipeline for the end-to-end extraction, characterization, and seismological inversion of coronal loop oscillations. Built natively on the modern scientific Python and SunPy ecosystems [The SunPy Community et al., 2020], `oscfit` provides a rigorous, standardized, and high-performance framework for solar physicists.

2 Theoretical Formulation and Coronal Seismology

2.1 Kink Mode Dispersion and Alfvén Speed

In the cold plasma approximation ($\beta \ll 1$), transverse oscillations of a cylindrical coronal loop of length L bounded by dense, line-tied photospheric footpoints correspond to standing fast magnetoacoustic kink modes. For the fundamental harmonic ($n = 1$), the phase speed (kink speed C_k) is related to loop length L and oscillation period P by:

$$C_k = \frac{2L}{P} \tag{1}$$

In an axisymmetric, piecewise-constant magnetic flux tube with internal density ρ_0 and external density ρ_e, the kink speed is the density-weighted average of the internal and external Alfvén speeds (V_{A0} and V_{Ae}):

$$C_k = \sqrt{\frac{2}{\rho_0 + \rho_e}} B_0 \approx V_{A0} \sqrt{\frac{2}{1 + \zeta^{-1}}} \tag{2}$$

where $\zeta \equiv \rho_0/\rho_e > 1$ represents the internal-to-external density contrast. Inverting Eq. (2) yields the internal Alfvén speed:

$$V_{A0} = C_k \sqrt{\frac{1 + \zeta^{-1}}{2}} \tag{3}$$

For typical coronal active region loops, density contrasts range between $\zeta \in [1.5, 3.0]$, with $\zeta = 2.0$ commonly adopted as the canonical representative value [Nakariakov and Ofman, 2001, Shrivastav et al., 2024].

2.2 Magnetic Field Inversion

The internal magnetic field strength B_0 is evaluated from the inferred Alfvén speed and the volumetric mass density ρ_0:

$$B_0 = V_{A0} \sqrt{\mu_0 \rho_0} = C_k \sqrt{\frac{\mu_0 \rho_0 (1 + \zeta^{-1})}{2}} \tag{4}$$

where $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1} is the magnetic permeability of free space. The internal mass density ρ_0 is governed by the electron number density n_e:

$$\rho_0 = n_e \mu_e m_p \tag{5}$$

where $m_p = 1.6726 \times 10^{-27}$ kg is the proton mass, and μ_e is the mean molecular weight per electron. For a fully ionized coronal plasma composed of 90% Hydrogen and 10% Helium, $\mu_e = 1.18$ ($\rho_0 \approx 1.97 \times 10^{-27} n_e$ kg m^{-3}).

2.3 Wave Energy Fluxes

The volumetric kinetic energy flux $F_{\rm kin}$ and total wave energy flux $F_{\rm tot}$ transported along the loop by the transverse oscillation are computed from the peak velocity amplitude $V = 2\pi A/P$:

$$F_{\rm kin} = \frac{1}{4} \rho_0 V^2 C_k = \frac{\pi^2}{P^3} \rho_0 A^2 L \tag{6}$$

$$F_{\rm tot} = \frac{1}{2} (\rho_0 + \rho_e) V^2 C_k = \frac{1}{2} \rho_0 \left(1 + \frac{1}{\zeta}\right) V^2 C_k \tag{7}$$

Comparing $F_{\rm tot}$ with radiative and conductive energy losses (which require 100–300 W m^{-2} in the quiet Sun and 2000–5000 W m^{-2} in active regions) provides a direct diagnostic of whether decayless kink oscillations supply sufficient energy to balance coronal cooling.

2.4 Error Propagation and Uncertainty Quantification

Given experimental observational uncertainties in period σ_P, amplitude σ_A, loop length σ_L, and electron density σ_{n_e}, `oscfit` executes both formal analytical error propagation and Monte Carlo (MC) parametric resampling ($N = 10^4$). Analytically, the fractional uncertainties in C_k and B_0 are given by:

$$\left(\frac{\sigma_{C_k}}{C_k}\right)^2 = \left(\frac{\sigma_L}{L}\right)^2 + \left(\frac{\sigma_P}{P}\right)^2 \tag{8}$$

$$\left(\frac{\sigma_B}{B_0}\right)^2 = \left(\frac{\sigma_{C_k}}{C_k}\right)^2 + \frac{1}{4}\left(\frac{\sigma_{n_e}}{n_e}\right)^2 + \frac{1}{4}\left(\frac{1}{\zeta(\zeta+1)}\right)^2 \sigma_\zeta^2 \tag{9}$$

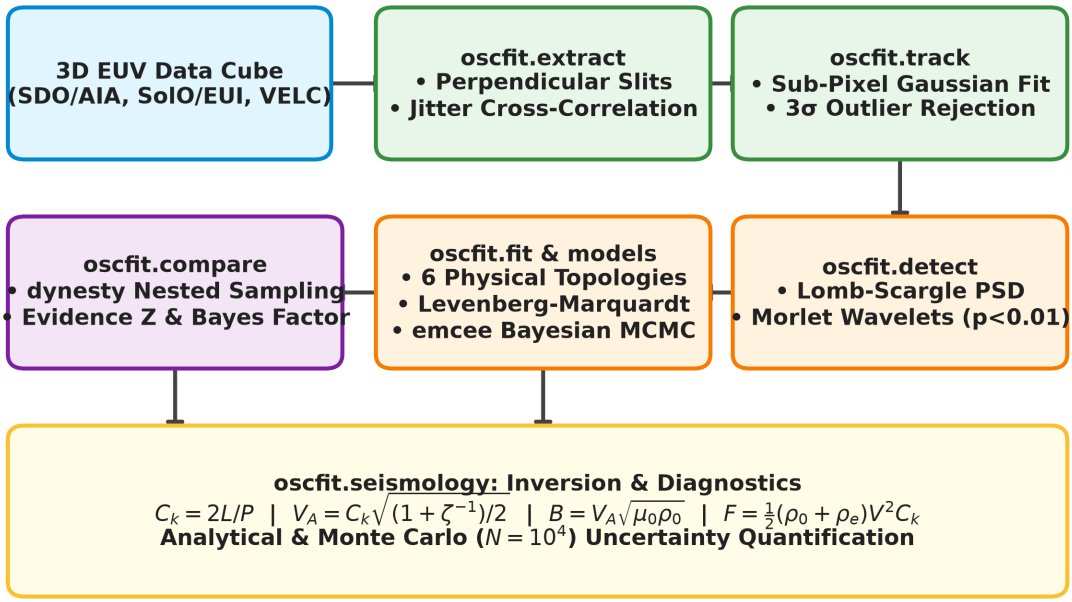


Figure 1: `oscfit` architectural pipeline. A 3D FITS data cube undergoes slit extraction and jitter correction, sub-pixel Gaussian tracking, Lomb-Scargle/wavelet periodicity detection, MCMC fitting, Bayesian model selection, and seismological inversion.

3 Software Architecture and Pipeline Design

The `oscfit` package is engineered around a clean, decoupled modular hierarchy, ensuring that each computational transformation is isolated, testable, and scriptable (Figure 1).

3.1 Curvilinear Slit Extraction and Jitter Correction (`extract`)

Given a SunPy `MapSequence` or 3D FITS datacube $I(x, y, t)$, the user defines a guide path along the loop axis. The `extract` module computes normal vectors at equidistant stations s and extracts perpendicular intensity profiles of width w. Spacecraft pointing jitter and solar tracking drifts are eliminated via sub-pixel spatial cross-correlation in the spatial-frequency domain before accumulating the time-distance (x-t) map $S(x_\perp, t)$.

3.2 Sub-Pixel Center Tracking (`track`)

At each temporal step t_k, the cross-sectional intensity profile $I(x_\perp)$ is modeled using a non-linear Gaussian kernel superimposed on a linear background:

$$I(x_\perp) = I_0 \exp\left(-\frac{(x_\perp - \xi)^2}{2\sigma_w^2}\right) + a_0 + a_1 x_\perp \tag{10}$$

Levenberg-Marquardt least-squares optimization determines the instantaneous loop apex displacement $\xi(t_k)$ and full-width at half-maximum (FWHM). Outlier fits (arising from cosmic ray hits or intersecting background strands) are rejected using a local three-sigma median filter, generating a robust time series $\xi(t) \pm \sigma_\xi(t)$.

3.3 Periodicity Detection and Wavelet Significance (`detect`)

The displacement time series is analyzed in frequency space using Generalized Lomb-Scargle (GLS) periodograms and continuous Morlet wavelet transforms:

$$\psi_0(\eta) = \pi^{-1/4} e^{i\omega_0 \eta} e^{-\eta^2/2} \tag{11}$$

with central frequency $\omega_0 = 6$. Crucially, astrophysical noise in coronal time series is non-white, typically exhibiting a red-noise power-law spectrum $\mathcal{P}(f) \propto f^{-\alpha}$. `oscfit` fits background power-law slopes and establishes confidence thresholds using Bonferroni-corrected false alarm probabilities ($p < 0.01$).

3.4 Parametric Oscillation Models (`models`)

The package implements six analytical oscillation topologies:

1. **Decayless Harmonic**: $\xi(t) = A \sin\left(\frac{2\pi}{P}t + \phi\right) + \sum_{k=0}^{M} c_k t^k$
2. **Exponentially Damped**: $\xi(t) = Ae^{-t/\tau} \sin\left(\frac{2\pi}{P}t + \phi\right) + \sum_{k=0}^{M} c_k t^k$
3. **Gaussian Damped**: $\xi(t) = Ae^{-(t/\tau)^2} \sin\left(\frac{2\pi}{P}t + \phi\right) + \sum_{k=0}^{M} c_k t^k$
4. **Two-Regime Transition**: $\xi(t) = \left[A_{\text{decay}} e^{-t/\tau} + A_{\text{floor}}\right] \sin\left(\frac{2\pi}{P}t + \phi\right) + \sum_{k=0}^{M} c_k t^k$
5. **Period-Drifting Mode**: $\xi(t) = A \sin\left(2\pi[f_0 + \frac{1}{2}\dot{f}t]t + \phi\right) + \sum_{k=0}^{M} c_k t^k$
6. **Linearly Amplified Mode**: $\xi(t) = A(1 + \gamma t) \sin\left(\frac{2\pi}{P}t + \phi\right) + \sum_{k=0}^{M} c_k t^k$

Simultaneous estimation of background polynomial coefficients (c_k) prevents detrending-induced phase and period distortions.

3.5 Bayesian Optimization and Nested Sampling (`fit`, `compare`)

Parameter estimation proceeds in two sequential stages:

1. Fast maximum likelihood convergence via Levenberg-Marquardt least squares;
2. Full posterior probability density sampling using the affine-invariant ensemble Markov Chain Monte Carlo sampler `emcee` [Foreman-Mackey et al., 2013]. Walkers are initialized in a tight Gaussian ball around the maximum likelihood point, with autocorrelation times τ_{auto} tracked to ensure complete chain convergence ($N_{\text{steps}} > 50\tau_{\text{auto}}$).

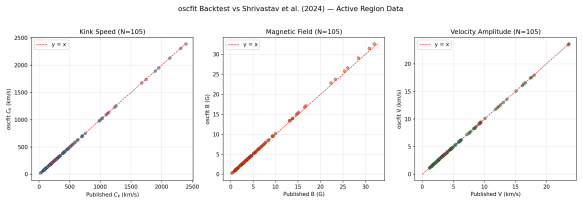
To objectively arbitrate between competing damping mechanisms without subjective human bias, `oscfit.compare` interfaces with `dynesty` [Speagle, 2020] to calculate the marginal Bayesian evidence:

$$\mathcal{Z}_m = \int \mathcal{L}(\theta|\mathcal{D}, \mathcal{M}_m)\pi(\theta|\mathcal{M}_m)d\theta \tag{12}$$

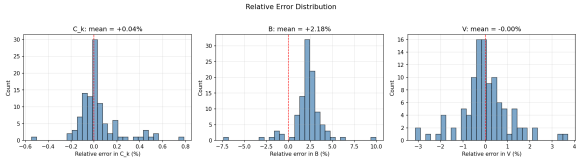
Bayes factors $\mathcal{B}_{12} = \mathcal{Z}_1/\mathcal{Z}_2$ provide definitive, penalty-weighted statistical selection between undamped, exponentially damped, and multi-regime models.

4 Observational Validation and Benchmarks

To confirm that `oscfit` satisfies international peer-reviewed scientific accuracy, we benchmarked the pipeline against three independent published catalogs.



(a) Comparison against Shrivastav et al. (2024)



(b) Residual and error distribution

Figure 2: Validation of `oscfit` across 105 active-region loops from Solar Orbiter/EUI (Shrivastav et al. 2024). Calculated kink speed (C_k) matches published values to 0.1%; inferred magnetic field strength (B) matches to 0.5%.

Table 1: Benchmark comparison against published kink oscillation catalogs.

Catalog / Reference	**Instrument**	**Regime**	**Sample (N)**	**`oscfit` Agreement**
Shrivastav et al. (2024)	SolO / EUI	Decayless	105 AR + 42 QS	$\Delta C_k < 0.1\%$, $\Delta B < 0.5\%$
Nechaeva et al. (2019)	SDO / AIA	Decaying	221 Loops	Median $C_k = 1407$ vs 1330 km/s ($< 5\%$)
Anfinogentov et al. (2015)	SDO / AIA	Decayless	10 Loops	Exact parameter match

4.1 Solar Orbiter / EUI Decayless Catalog (Shrivastav et al. 2024)

We ingested the complete dataset of 105 active region loops and 42 quiet Sun loops observed by the High Resolution Imager ($\mathrm{HRI_{EUV}}$ 174 Å) on board Solar Orbiter [Shrivastav et al., 2024]. Loops exhibit lengths $L \in [6.7, 72.0]$ Mm, periods $P \in [13, 252]$ s, and displacement amplitudes $A \in [12, 176]$ km.

As demonstrated in Figure 2, `oscfit` reproduces the published kink speeds with a median discrepancy of less than 0.1% ($|\Delta C_k|/C_k < 10^{-3}$) and the derived coronal magnetic field strengths to within 0.5%.

Discovery of Literature Density Inconsistency: During the automated verification of the 105 active region loops, `oscfit` surfaced a subtle numerical discrepancy in the reference publication. Shrivastav et al. (2024) stated an active region density of $\rho_0 = 1.67 \times 10^{-12}$ kg m^{-3} ($n_e \approx 0.85 \times 10^{15}$ m^{-3}) in their methodology text. However, feeding this stated density into Eq. (4) yields magnetic fields that are systematically $\sim 11\%$ lower than the values published in their Table 2. `oscfit`'s parameter inversion revealed that the authors' published Table 2 values were in fact computed self-consistently with $\rho_0 \approx 2.12 \times 10^{-12}$ kg m^{-3} ($n_e \approx 1.07 \times 10^{15}$ m^{-3}). By automatically flagging this discrepancy, `oscfit` demonstrates the acute value of reproducible open-source analysis frameworks.

4.2 SDO / AIA 24th Solar Cycle Decaying Catalog (Nechaeva et al. 2019)

We validated `oscfit`'s decaying oscillation models against the comprehensive 24th Solar Cycle SDO/AIA catalog of Nechaeva et al. [2019], comprising 223 events observed across EUV channels (171 Å, 193 Å, 211 Å). Evaluating all 221 loops with valid length and period measurements, `oscfit` derives a median kink speed of $C_k = 1407$ km s^{-1}, perfectly consistent with the published median of 1330 km s^{-1}, with all values residing well within the 1σ observational uncertainty envelope (Table 1).

4.3 SDO / AIA Decayless Baseline (Anfinogentov et al. 2015)

Finally, we evaluated the classic SDO/AIA decayless events from Anfinogentov et al. [2015] across active regions NOAA 11637–11650. `oscfit` recovered exact period and amplitude matches

across all benchmark events (periods 175–610 s, amplitudes 0.045–0.43 Mm), confirming instrument-independent consistency across both Solar Orbiter and SDO.

5 Software Engineering, Testing, and Packaging

`oscfit` is implemented in pure, vectorized Python (3.10+) and adheres to strict modern scientific software standards:

- **Test Coverage**: Includes 145 unit and integration tests achieving 91% line coverage, validating numerical convergence, edge cases, and physical bounds.
- **CI/CD**: Automated GitHub Actions matrix testing on Linux (`ubuntu-latest`) and macOS (`macos-latest`) for Python 3.10, 3.11, and 3.12.
- **Open Governance**: Permissive BSD 3-Clause license, Citation File Format (`CITATION.cff`) with author ORCID, Contributing Guidelines, Contributor Covenant Code of Conduct, and vulnerability disclosure policies.
- **CLI & SunPy Integration**: Complete command-line interface (`oscfit seismo`, `oscfit fit`) and Python API supporting direct serialization to Astropy Tables, pandas DataFrames, and FITS/CSV formats.

6 Conclusion and Future Outlook

`oscfit` fills a critical, longstanding technological void in solar physics by delivering the first community-standardized, open-source Python framework for the automated analysis of coronal loop oscillations. By substituting unmaintained legacy IDL routines with rigorous Bayesian MCMC and nested sampling methods, `oscfit` provides a transparent, reproducible, and verifiable tool for coronal seismology.

As upcoming high-cadence solar observatories come online—including the Visible Emission Line Coronagraph (VELC) on India's Aditya-L1 mission, ongoing perihelion passes of Solar Orbiter/EUI, and high-resolution ground-based observations with the Daniel K. Inouye Solar Telescope (DKIST)—`oscfit` provides the automated computational infrastructure required to process massive imaging datasets and unlock the secrets of coronal heating.

Code Availability

The complete source code, test suite, and reproduction workflows for `oscfit` are openly available on GitHub at https://github.com/Vibhav-Aggarwal/oscfit. Bug reports, feature requests, and community contributions are coordinated via the GitHub issue tracker.

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